Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which of the following compounds will not show the test of nitrogen?

(A) $CH_3CH_2NH_2$

(B) $C_6H_5NH_2$

(C) NH_2OH

(D) NH_2CONH_2

Q.2

The composition of species showing purple colour in the test of S when sodium nitroprusside is added to the Lassaigne solution is

(A) $[Fe(CN)_5NO]^{2-}$

(B) $[Fe(CN)_5NOS]^{4}$

(C) $[Fe(CN)_5(NO)S]^{2-}$

(D) $[Fe(CN)_4NOS]^{3-}$

Q.3

In the Fehling's solution test for an aldehyde, the red precipitates is due to the formation of

(A) cupric nitrate

(B) copper

(C) cupric oxide

(D) cuprous oxide

Q.4

Which of the following compounds will show iodoform test?

(A) C_6H_5CHO

(B) $CH_3CH_2COCH_2CH_3$

(C) CH_3CHO

(D) CH_3COOH

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Q.5

Molisch test is meant for

(A) alcohols

(B) carbohydrates

(C) aldehydes

(D) haloalkanes

Q.6

Phenols does not react with

(A) $NaHCO_3$

(B) Na_2CO_3

(C) $NaOH$

(D) Br_2

Q.7

Carbylamine test is for the detection of

(A) primary amines

(B) secondary amines

(C) tertiary amines

(D) quaterly ammonium salt

Q.8

Which of the following compounds liberate N_2 on heating?

(A) $(NH_4)_2\,Cr_2O_7$

(B) $Cu\,(NO_3)_2$

(C) $AgNO_3$

(D) NH_4NO_3

Q.9

Which of the following compounds shows yellow appearance on heating and becomes white when cooled after heating?

(A) $CuCO_3$

(B) PbO_2

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(C) $Pb(NO_3)_2$

(D) $ZnCO_3$

Q.10

Which of the following salts gives a mixture of SO_2 and SO_3 on heating?

(A) $CuSO_4$

(B) $FeSO_4$

(C) $BaSO_4$

(D) Na_2SO_4

Q.11

Which of the following compounds gives white precipitate turning to brown and ultimately black when treated with silver nitrate solution?

(A) KNO_3

(B) K_2SO_4

(C) $Na_2\ S_2O_3$

(D) Na_2SO_3

Q.12

Silver chloride is soluble in ammonium hydroxide due to the formation of

(A) $AgNH_2$

(B) $[Ag(NH_3)_2]Cl$

(C) $Ag(NH_3)Cl$

(D) $[Ag(NH_3)_6]Cl$

Q.13

Heating of phosphate salt to about $50-60^{\circ}C$ with concentrated HNO_3 and ammonium molybdate produces

(A) yellow precipitates of $(NH_4)_3[P(Mo_3O_{10})_4]$

(B) red precipitates of $(NH_4)_3PO_4 \cdot 10MoO_3$

(C) gree precipitates of $(NH_4)_3PO_4 \cdot 6MoO_3$

(D) blue precipitates of $MO_3(PO_4)_2$

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Q.14

Which of the following pairs of compounds are precipitated in different group in the systematic analysis of cations?

(A) Bi^{3+}, As^{3+}

(B) As^{3+}, Sn^{2+}

(C) Cu^{2+}, Zn^{2+}

(D) Ba^{2+}, Sr^{2+}

Q.15

Nessler's reagent is

(A) ammoniacal solution of K_2HgI_4

(B) acidic solution of K_2HgI_4

(C) alkaline solution of K_2HgI_4

(D) aqueous solution of KI

Q.16

The composition of white precipitate when tartaric acid solution is added to a solution containing K^+ is

(A)
$$\begin{array}{l}CH(OK)COOK\\|\\CH(OH)COOK\end{array}$$

(B)
$$\begin{array}{l}CH(OH)COOH\\|\\CH(OH)COOK\end{array}$$

(C)
$$\begin{array}{l}CH(OH)COOH\\|\\CH(OK)COOH\end{array}$$

(D)
$$\begin{array}{l}CH(OH)COOK\\|\\CH(OH)COOK\end{array}$$

Q.17

The composition of yellow precipitate when a solution of ammonium molybdate is added to hot solution of arsenic salt is

(A) $(NH_4)_3\,AsO_4 \cdot 12MoO_2$

(B) $(NH_4)_3\,AsO_4 \cdot 6MoO_2$

(C) $(NH_4)_3\,AsO_4 \cdot 4MoO_2$

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(D) $(NH_4)_3\,AsO_4 \cdot MoO_2$

Q.18

Which of the sulphides of Group IV is white in colour?

(A) Co^{2+}

(B) Ni^{2+}

(C) Mn^{2+}

(D) Zn^{2+}

Q.19

Which of the following compounds forms white precipitate with $K4Fe(CN)_6$?

(A) Cu^{2+}

(B) NH_4^+

(C) Zn^{2+}

(D) Ca^{2+}

Q.20

A metallic nitrate gives black precipitates with KI solution. On adding excess of KI, an orange-coloured solution is obtained. The ion giving these observations is

(A) Hg^{2+}

(B) Sn^{2+}

(C) Bi^{3+}

(D) Cd^{2+}

Q.21

Match the organic compounds in column - I with the Lassaigne's test results in column - II appropriately: \begin{array}{l} \ \begin{array}{|l|l|} \ \hline & \text { Column - I } & & \text { Column - II } \\ \ \hline \text { (A) } & \text { Aniline } & \text { (i) } & \text { Red color with } \mathrm{FeCl}_{3} \\ \ \hline \text { (B) } & \begin{array}{l} \ \text { Benzene sulfonic } \\ \ \text { acid } \ \end{array} & \text { (ii) } & \text { Violet color with sodium nitroprusside } \\ \ \hline \text { (C) } & \text { Thiourea } & \text { (iii) } & \mathrm{Blue} \text { color with hot and acidic solution of } \\ \ & & & \mathrm{FeSO}_{4} \ \end{array}\\ \ \hline\\ \ \end{array}

(A) A$-$ (ii); B$-$ (i); C$-$ (iii)

(B) A$-$ (iii); B$-$ (ii); C$-$ (i)

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(C) $\mathrm{A}-$ (ii); $\mathrm{B}-$ (iii); $\mathrm{C}-$ (i)

(D) $\mathrm{A}-$ (iii); $\mathrm{B}-$ (i); $\mathrm{C}-$ (ii)

Q.22

Which of the following statements is correct?

(A) On boiling a solution having $\mathrm{K}^{+}, \mathrm{Ca}^{2+}$ and HCO_3^{-} ions we get a precipitate of $\mathrm{K}_2\mathrm{Ca}\left(\mathrm{CO}_3\right)_2$

(B) Manganese salts give a violet borax bead test in the reducing flame

(C) From a mixed precipitate of AgCl and AgI, ammonia solution dissolves only AgCl

(D) Ferric ions give a deep green precipitate on adding potassium ferrocyanide solution

Q.23

Excess of KI reacts with CuSO_4 solution and then $\mathrm{Na}_2\mathrm{S}_2\mathrm{O}_3$ solution is added to it. Which of the statements is incorrect for this reaction?

(A) $\mathrm{Cu}_2\mathrm{I}_2$ is formed

(B) CuI_2 is formed

(C) $\mathrm{Na}_2\mathrm{S}_2\mathrm{O}_3$ is oxidised

(D) evolved I_2 is reduced

Q.24

$29.5\ \mathrm{mg}$ of an organic compound containing nitrogen was digested according to Kjeldahl's method and the evolved ammonia was absorbed in $20\ \mathrm{mL}$ of $0.1\ \mathrm{M}\ \mathrm{HCl}$ solution. The excess of the acid required $15\ \mathrm{mL}$ of $0.1\ \mathrm{M}\ \mathrm{NaOH}$ solution for complete neutralization. The percentage of nitrogen in the compound is

(A) 29.5

(B) 59.0

(C) 47.4

(D) 23.7

Q.25

The products obtained on heating LiNO_3 will be

(A) $\mathrm{LiNO}_2 + \mathrm{O}_2$

(B) $\mathrm{Li}_2\mathrm{O} + \mathrm{NO}_2 + \mathrm{O}_2$

(C) $\mathrm{Li}_3\mathrm{N} + \mathrm{O}_2$

(D) $\mathrm{Li}_2\mathrm{O} + \mathrm{NO} + \mathrm{O}_2$

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Q.1 Answer:

NH_2OH

Solution:

hydroxylamine (NH_2OH) doesn't give a positive test for N since it does not contain C. So $CN^{\ominus}$ ion is not obtained which is an essential condition for the test for N.

Q.2 Answer:

$[Fe(CN)_5NOS]^4$

Solution:

The appearance of a deep violet color upon addition of a few drops of sodium nitroprusside to SFE indicates the presence of sulfur.

Detection of sulfur: formation of violet color with nitroprusside

$$\underset{}{S^{2-}} + \underset{\text{Nitroprusside ion}}{[Fe(CN)_5NO]^{2-}} \longrightarrow \underset{\text{violet color}}{[Fe(CN)_5NOS]^{4-}}$$

Q.3 Answer:

cuprous oxide

Solution:

The reaction between copper (II) ions and aldehyde in Fehling's solution is represented as;

$$RCHO + 2Cu^{2+} + 5OH^- \rightarrow RCOO^- + Cu_2O + 3H_2O$$

When tartrate is added, the reaction can be written as:

$$RCHO + 2Cu\left(C_4H_4O_6\right)_2{}^{2-} + 5OH^- \rightarrow RCOO^- + Cu_2O + 4C_4H_4O_6^{2-} + 3H_2O$$

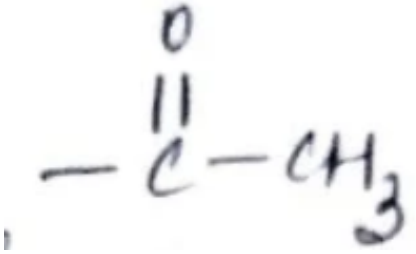
When the redox reaction is completed, the copper (II) ions are reduced to Copper (I) oxide, which forms a red precipitate and is insoluble in water. A positive test result is indicated by the presence of this red precipitate.

Q.4 Answer:

CH_3CHO

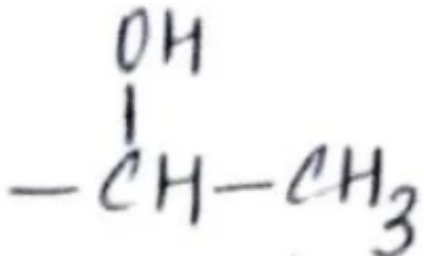
Solution:

A compound which contains either

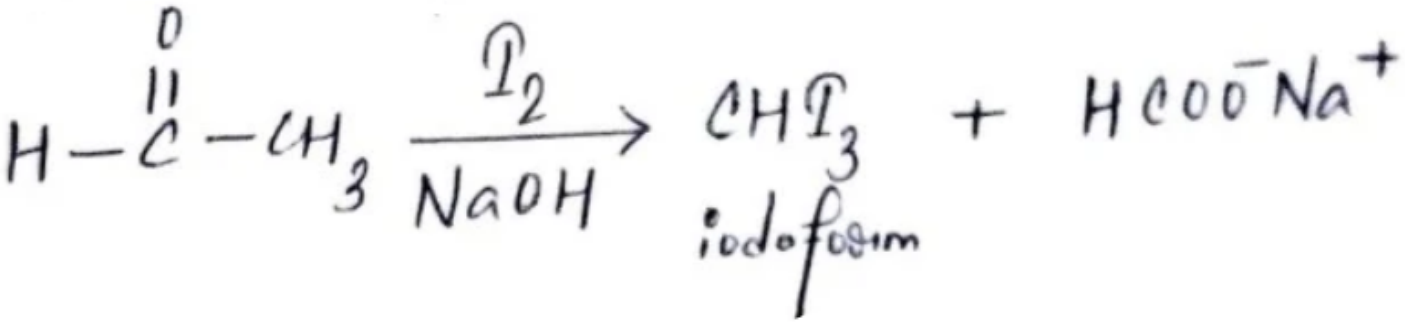


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or



aroup gives positive idoform test. So, CH_3CHO **will show iodoform test.**



Q.5 Answer:

carbohydrates

Solution:

Molisch's test is a sensitive chemical test, named after Austrian botanist Hans Molisch, for the presence of carbohydrates, based on the dehydration of the carbohydrate by sulfuric acid or hydrochloric acid to produce an aldehyde, which condenses with two molecules of a phenol (usually α**-naphthol, though other phenols such as resorcinol and thymol also give colored products), resulting in a violet ring.**

Q.6 Answer:

$NaHCO_3$

Solution:

Phenols are weakly acidic and bicarbonate ions are not strong enough to take the H^+ **from phenols. Hence, it behaves differently from other acids in that it does not react with sodium bicarbonate. Option (a) is correct.**

Q.7 Answer:

primary amines

Solution:

Only primary amines can give carbylamine test.

$RNH_2 + CHCl_3 + KOH \rightarrow RNC$

$Ar - NH_2 + CHCl_3 + KOH \rightarrow Ar - NC$

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Here, R - may be an alkyl or an aryl group.

Secondary and tertiary amines don't give carbylamine reaction test.

Q.8 Answer:

$(\mathrm{NH_4})_2\,\mathrm{Cr_2O_7}$

Solution:

$(\mathrm{NH_4})_2\,\mathrm{Cr_2O_7}$ liberates N_2 on heating

$(\mathrm{NH_4})_2\,\mathrm{Cr_2O_7} \xrightarrow{\Delta} N_2 \uparrow + \mathrm{Cr_2O_3} + 4\mathrm{H_2O}$

Q.9 Answer:

$\mathrm{ZnCO_3}$

Solution:

Zinc salts on heating appear yellow and become white on cooling that is why $\mathrm{Z_nCO_3}$ (containing $\mathrm{Zn^{2+}}$) turns yellow when heated and white when cooled.

Q.10 Answer:

$\mathrm{FeSO_4}$

Solution:

On heating, ferrous sulphate crystals lose water and anhydrous ferrous sulphate $(\mathrm{FeSO_4})$ is foumed. So their colour changes from light green to white. On further heating ; anhydrous ferrous sulphate decomposes to form ferric oxide $(\mathrm{Fe_2O_3})$ sulphur dioxide, $(\mathrm{SO_2})$, and sulphur trioxide, $(\mathrm{SO_3})$.

The reaction is as follows:

$2\mathrm{FeSO_4} \xrightarrow{\Delta} \mathrm{Fe_2O_3} + \mathrm{SO_2} + \mathrm{SO_3}$

Q.11 Answer:

$\mathrm{Na_2\,S_2O_3}$

Solution:

On reaction of a hyposolution i.e. sodium thiosulphate with silver nitrate, silver thiosulphate is formed that goes through a colour change i.e. from white ppt to black ppt

$\mathrm{Na_2S_2O_3} + 2\mathrm{AgNO_3} \rightarrow \mathrm{Ag_2S_2O_3} + 2\mathrm{NaNO_3}$

$\mathrm{Ag_2NO_3} + \mathrm{H_2O} \rightarrow \mathrm{Ag_2S} + \mathrm{H_2SO_4}$

Q.12 Answer:

$[\mathrm{Ag\,(NH_3)_2}]\,\mathrm{Cl}$

Solution:

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$AgCl + 2NH_3 \longrightarrow [Ag(NH_3)_2]Cl$

White ppt. of $AgCl$ dissolves in NH_3 due to the formation of complex diammine silver chloride.

Q.13 Answer:

yellow precipitates of $(NH_4)_3[P(Mo_3O_{10})_4]$

Solution:

When ammonium molybdate is treated with a solution containing phosphate a canary yellow crystalline precipitate of ammonium phosphomolybdate (ammonium dodecamolybdatophosphate), $(NH_4)_3[PMo_{12}O_{40}]$ or $(NH_4)_3[P(Mo_3O_{10})_4]$ is formed.

In the compound formed, the Mo_3O_{10} group replaces each oxygen atom in phosphate.

$$PO_4^{3-} + 3NH_4^+ + 12MoO_4^{2-} + 24H^+ \rightarrow \underset{\text{Canary-Yellow ppt}}{(NH_4)_3[P(Mo_3O_{10})_4]\downarrow} + 12H_2O$$

Q.14 Answer:

Cu^{2+}, Zn^{2+}

Solution:

Cu^{2+} gives precipitate in group II duing qualitative analysis while Zn^{2+} gives precipitate in group III

for Cu^{2+}, $Cu^{2+} + H_2S \longrightarrow CuS\downarrow(black) + 2H^+$

for Zn^{2+}, $Zn^{2+} + H_2S \longrightarrow ZnS\downarrow(white) + 2H^+$

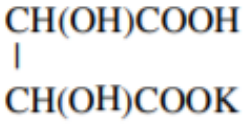
Q.15 Answer:

alkaline solution of K_2HgI_4

Solution:

Nessler's reagent is $K_2Hgl_4 + KOH$. It is used as a test for the presence of ammonia.

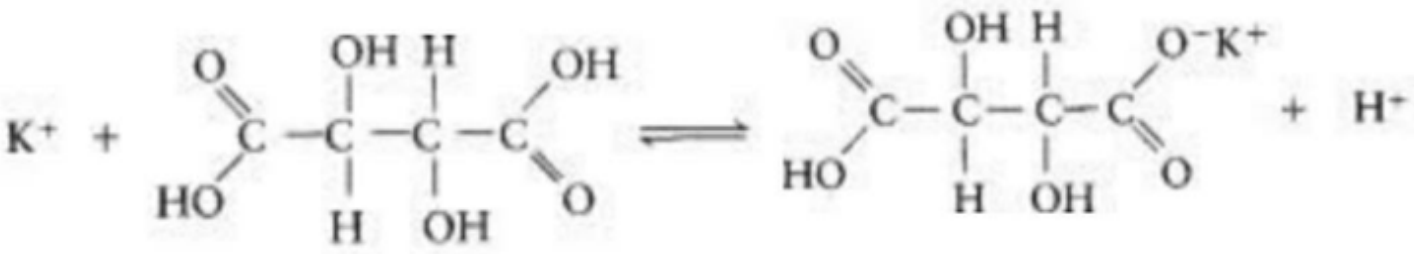
Q.16 Answer:



Solution:

When tartaric acid solution is added to a solution containing K^+, a white precipitate of potassium bitartarate is obtained.

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Q.17 Answer:

$(NH_4)_3\,AsO_4 \cdot 12MoO_2$

Solution:

$$H_3AsO_4 + 12(NH_4)_2.MoO_4 + 21HNO_3 \rightarrow \underset{\text{Ammonium arsenomolybdate}}{(NH_4)_3AsO_4.12MoO_3} \downarrow + 12H_2O + 21NH_4NO_3$$

Q.18 Answer:

Zn^{2+}

Solution:

ZnS — zinc sulphide white precipitate

CoS — cobalt sulphide black precipitate

MnS — manganese sulphide pink precipitate

NiS — nickel sulphide black precipitate

Q.19 Answer:

Zn^{2+}

Solution:

Zn^{2+} **reacts with potassium ferocyanide to form white ppt of zinc ferrocyanide.**

$Zn^{2+} + 2K^+ [Fe(CN)_6]^{4-} \rightarrow K_2Ca\,[Fe(CN)_6] \downarrow$ (**white ppt.**)

Q.20 Answer:

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Bi^{3+}

Solution:

$Bi^{3+} + KI \rightarrow BiI_3 \downarrow \text{ Sol.}$

Black

$Bi\left(NO_3\right)_3(aq) + 3KI(aq) \rightarrow BiI_3(s) + 3KNO_3(aq)$

Q.21 Answer:

A – (iii); B – (ii); C – (i)

Solution:

Lassaigne's test:

In a qualitative analysis of organic compounds, a test for the detection of nitrogen, sulfur, and halogens present in organic compounds is Lassaigne's test. In this test, the organic compound and a small piece of Na metal are heated in a fusion tube. The basic principle of this test is that Na converts the elements in organic compound into ionic form

(A) Aniline with Lassaigne's test: the aniline compound is boiled $FeSO_4$ with concentrated H_2SO_4, which will appear on Prussian blue color indicates the presence of nitrogen in the given organic compound.

(B) benzene sulfonic acid with Lassaigne's test: the given compound treated with sodium nitroprusside gives the appearance of violet color indicates the sulfur present in the benzene sulfonic acid.

(C) thiourea with Lassaigne's test: this compound on reacting with $FeCl_3/FeSO_4$ will give blood-red color.

Q.22 Answer:

From a mixed precipitate of $AgCl$ and AgI, ammonia solution dissolves only $AgCl$

Solution:

Between $AgCl$ and AgI, AgI is less soluble, hence ammonia can dissolve ppt. of $AgCl$ only due to formation of complex as given below: $AgCl + 2NH_3 \rightarrow [Ag(NH_3)_2]Cl$

Q.23 Answer:

CuI_2 is formed

Solution:

The reactions occurring are

$2CuSO_4 + 4KI \rightarrow Cu_2I_2 + I_2 + 2K_2SO_4; \quad 2Na_2S_2O_3 + I_2 \rightarrow Na_2S_4O_6 + 2NaI$

In these reactions, CuI_2 is not formed.

Q.24 Answer:

23.7

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Solution:

Amount of NaOH consumed = Amount of HCl unreacted

$n = VM = \left(15 \times 10^{-3}\ \mathrm{L}\right)\left(0.1\ \mathrm{mol\ L^{-1}}\right) = 15 \times 10^{-4}\ \mathrm{mol}$

Amount of $\mathrm{NH_3}$ evolved = Amount of HCl reacted

= Amount of HCl taken to start with - Amount of HCl unreacted

$= \left(20 \times 10^{-3}\ \mathrm{L}\right)\left(0.1\ \mathrm{mol\ L^{-1}}\right) - 15 \times 10^{-4}\ \mathrm{mol}$

$= 5 \times 10^{-4}\ \mathrm{mol}$

Mass of $\mathrm{NH_3}$ evolved $= nM = \left(5 \times 10^{-4}\ \mathrm{mol}\right)\left(17\ \mathrm{g\ mol^{-1}}\right) = 85 \times 10^{-4}\ \mathrm{g}$

Mass per cent of nitrogen in the compound is $\left(\frac{M_{\mathrm{N}}}{M_{\mathrm{NH_3}}}\right)\left(\frac{m_{\mathrm{NH_3}}}{m_{\mathrm{compound}}}\right) \times 100 =$

$\left(\frac{14\ \mathrm{g\ mol^{-1}}}{17\ \mathrm{g\ mol^{-1}}}\right)\left(\frac{85 \times 10^{-4}\ \mathrm{g}}{29.5 \times 10^{-3}\ \mathrm{g}}\right)(100) = 23.7$

Q.25 Answer:

$\mathrm{Li_2O + NO_2 + O_2}$

Solution:

$4\mathrm{LiNO_3} \longrightarrow 2\mathrm{Li_2O} + 4\mathrm{NO_2} + \mathrm{O_2}$

Lithium nitrate behaves differently from other alkali metal nitrates. Lithium nitrate on thermal decomposition produces metal oxide, $\mathrm{NO_2}$ and $\mathrm{O_2}$. While other alkali metal nitrate on heating produces metal nitrite and oxygen.

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